

CS 334 – Project Research Qs and Blog Post

Blog Post: <https://medium.com/@umerasghar6754/analyzing-factors-influencing-student-performance-a-data-driven-approach-360fc2495fc4>

Research Questions

1. Impact of Parental and Socio-Economic Factors on Exam Performance

- **Research Question:** What is the effect of parental involvement, access to resources, family income level, and parental education level on student exam performance (and, by proxy, overall student performance)?
- **Objective:** This question aims to explore how family support and socio-economic factors influence academic success. By analyzing these variables, we seek to understand the role of a student's home environment and resources in shaping educational outcomes. Insights could inform policies and practices to provide support where needed, addressing educational disparities.

2. Combined Influence of Academic, Behavioral, and Environmental Factors on Exam Scores

- **Research Question:** How do various factors, including hours studied, attendance, parental involvement, access to resources, extracurricular activities, sleep hours, previous academic performance, motivation level, internet access, tutoring sessions, family income, teacher quality, school type, peer influence, physical activity, learning disabilities, parental education level, distance from home, and gender, collectively impact students' exam scores (and by proxy, overall student performance)?
- **Objective:** This question broadens the scope to examine a holistic view of student performance, assessing how a combination of academic, behavioral, and environmental factors interact to affect outcomes. Understanding these interactions can help build predictive models for academic success and identify key areas for targeted interventions that support balanced student growth.